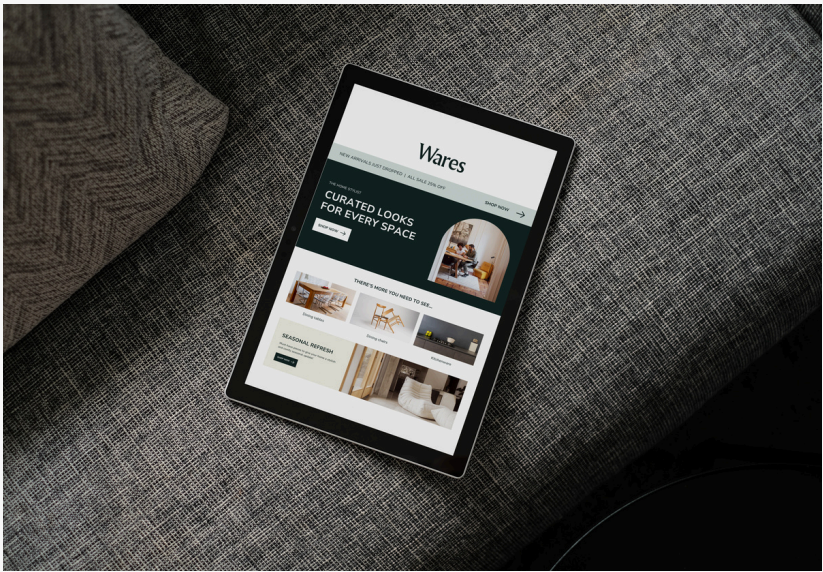


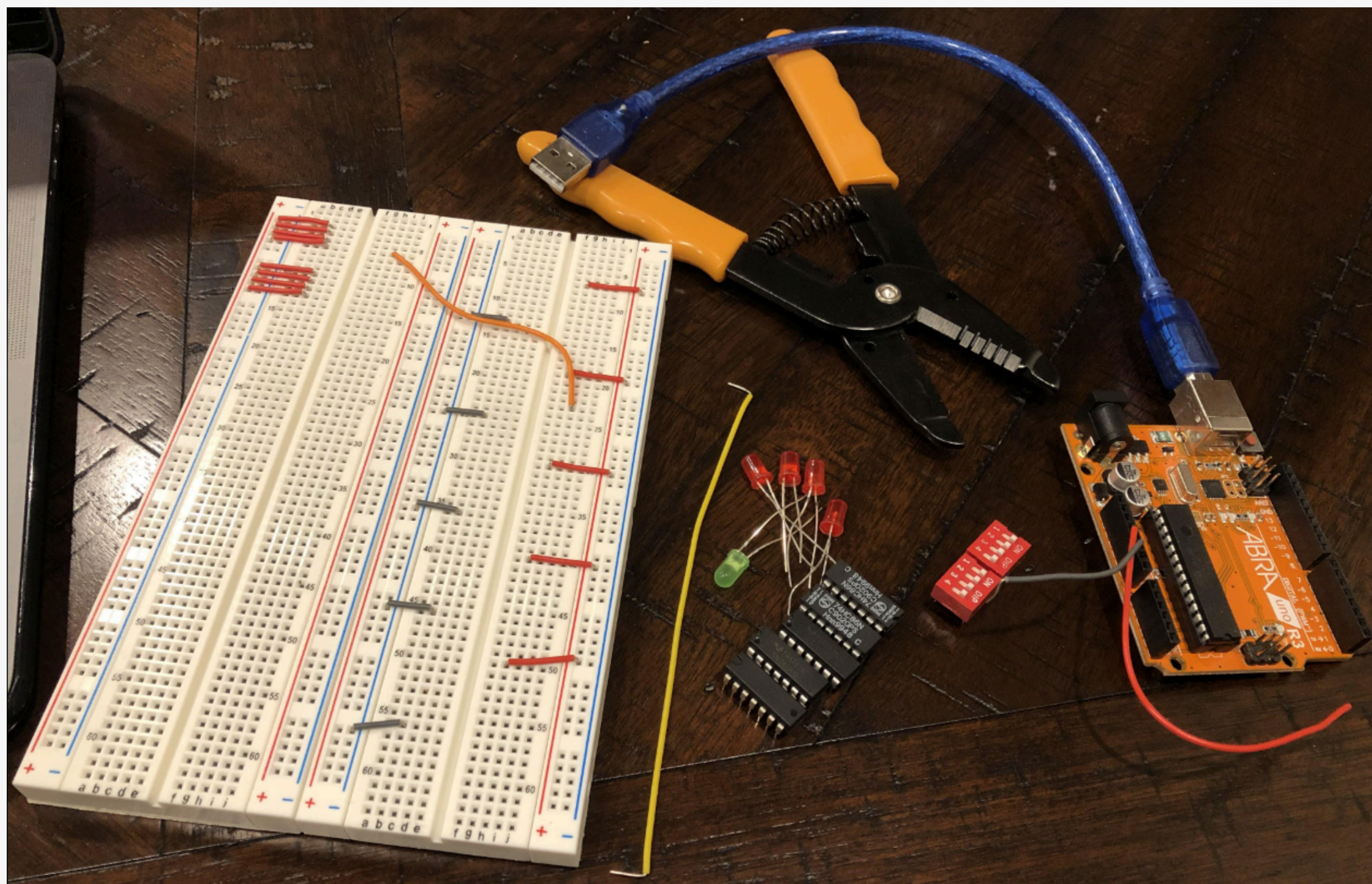
4-Bit Binary Adder



Overview

The 4-bit binary adder is implemented on an Arduino platform, capable of performing binary addition of two 4-bit numbers and displaying the result through digital outputs.





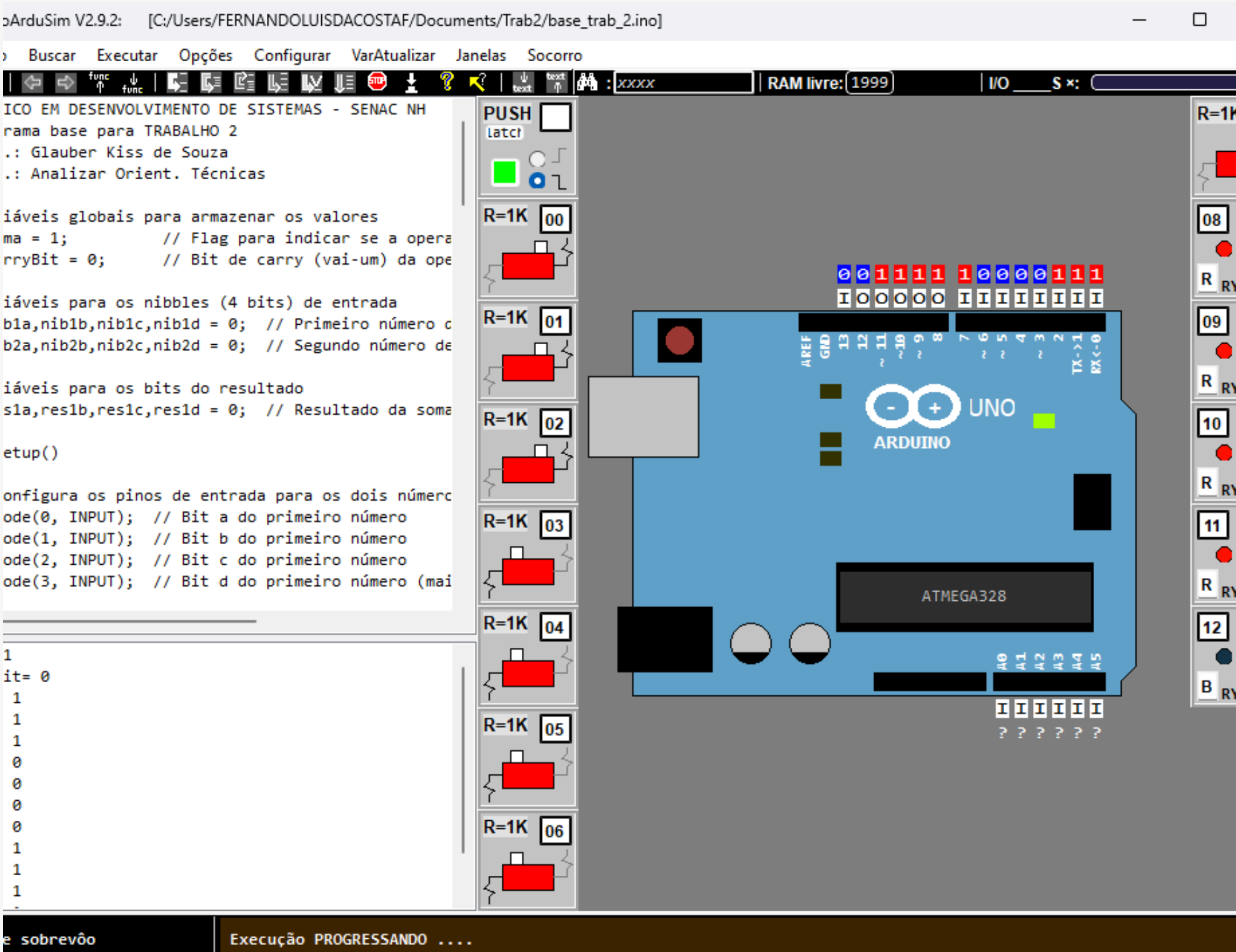
Key Features and Operation

Key Features

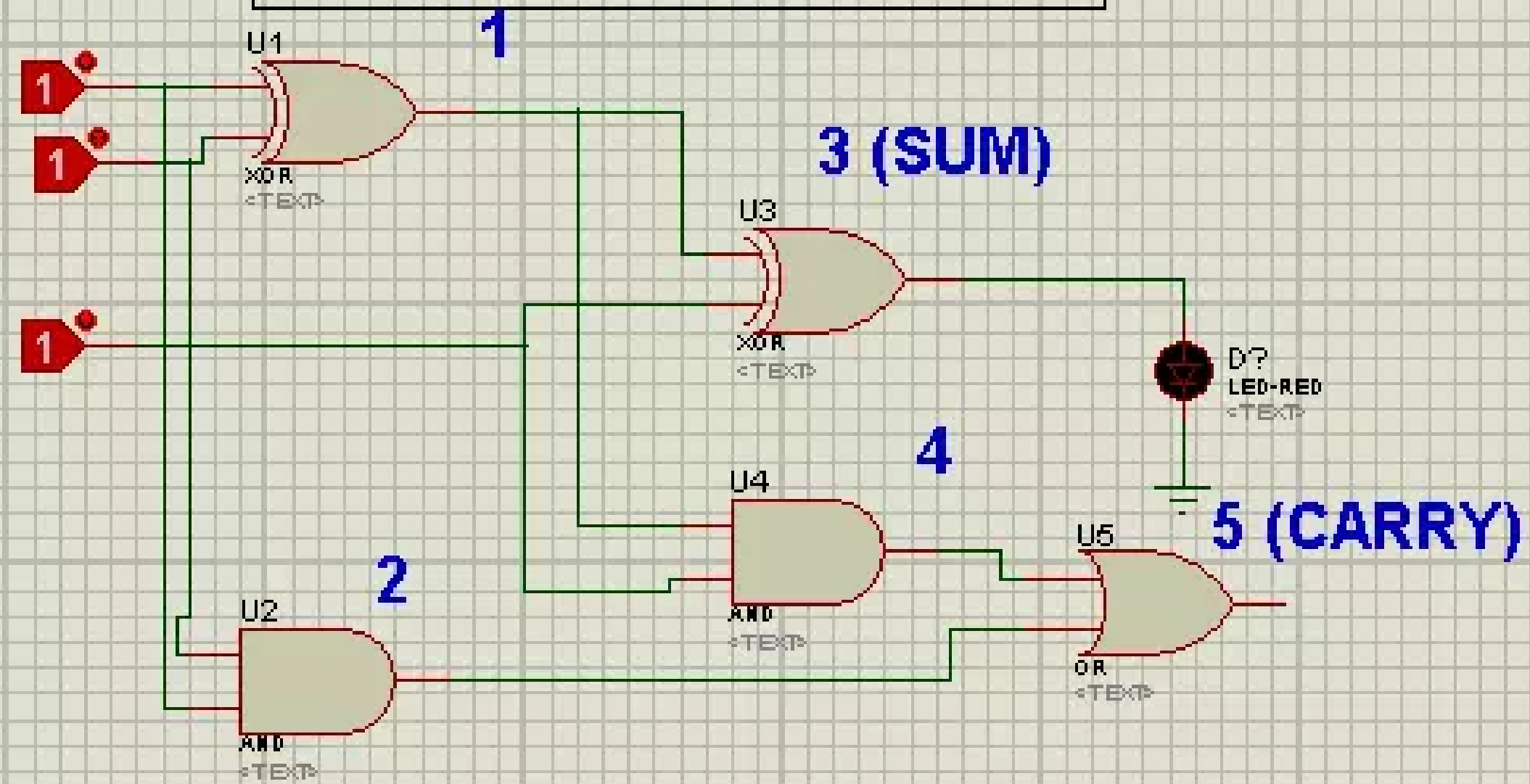
- Reads two 4-bit binary numbers from digital input pins
- Performs binary addition using full adder logic
- Displays 4-bit result and carry bit on output pins
- Real-time operation with continuous input monitoring

Operation

1. Input is provided through pins 0-7 (two 4-bit numbers)
2. System calculates the sum using bitwise operations
3. Result is displayed on pins 8-11 (4-bit result) and pin 12 (carry)
4. Process repeats continuously in the main loop



FOUR BITS FULL ADDER



Development Summary

Implemented Functionality

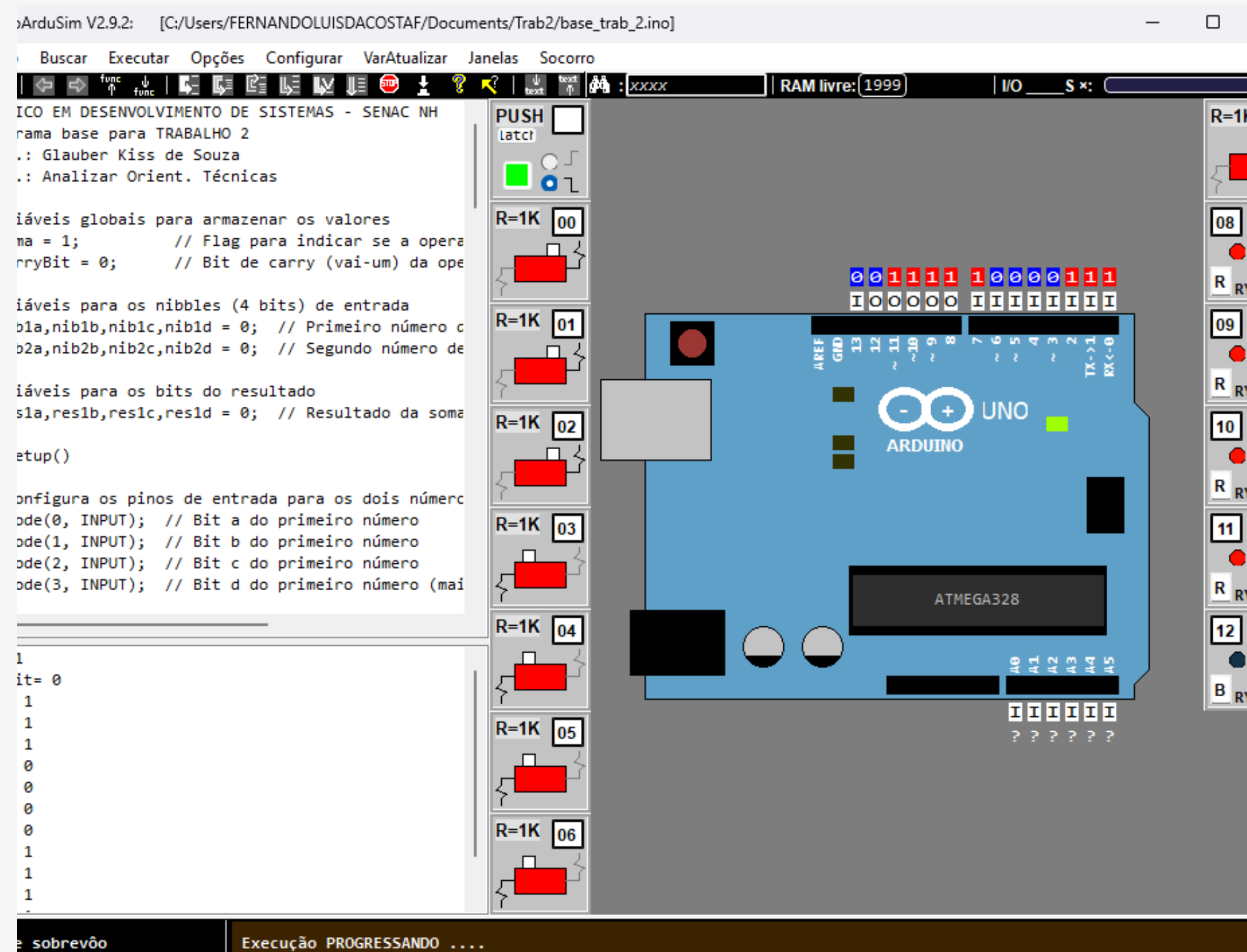
- Complete 4-bit binary adder circuit emulation
- Input reading from digital pins (8 input bits)
- Output writing to digital pins (5 output bits)
- Full adder implementation using boolean algebra:
 - Sum calculation: $S = A \oplus B \oplus C_{in}$
 - Carry calculation: $C_{out} = (A \cdot B) + (A \cdot C_{in}) + (B \cdot C_{in})$
- Continuous operation loop for real-time calculation

Technical Specifications

- Platform: Arduino Uno (compatible)
- Input: 8 digital pins (0-7)
- Output: 5 digital pins (8-12)
- Logic: Binary arithmetic using full adders
- Code: Written in Arduino C++

Educational Value

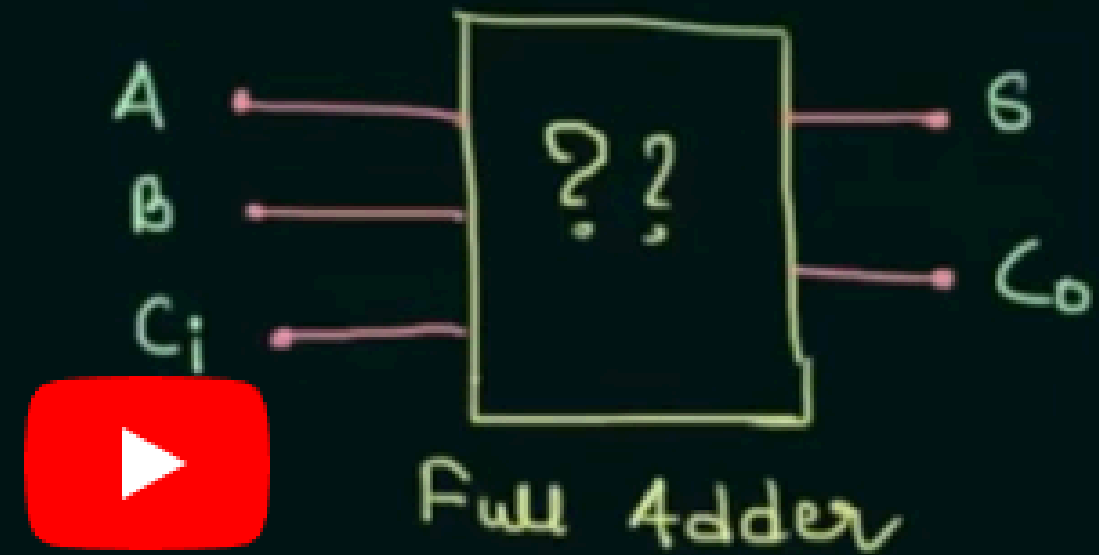
- Demonstrates fundamental digital logic concepts
- Illustrates binary arithmetic operations
- Shows hardware-software integration
- Provides foundation for more complex ALU designs



Full Adder

Full Adder

A	B	C_i	S	C_o
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1



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Digital Electronics

Future Development Activities

Short-term Enhancements

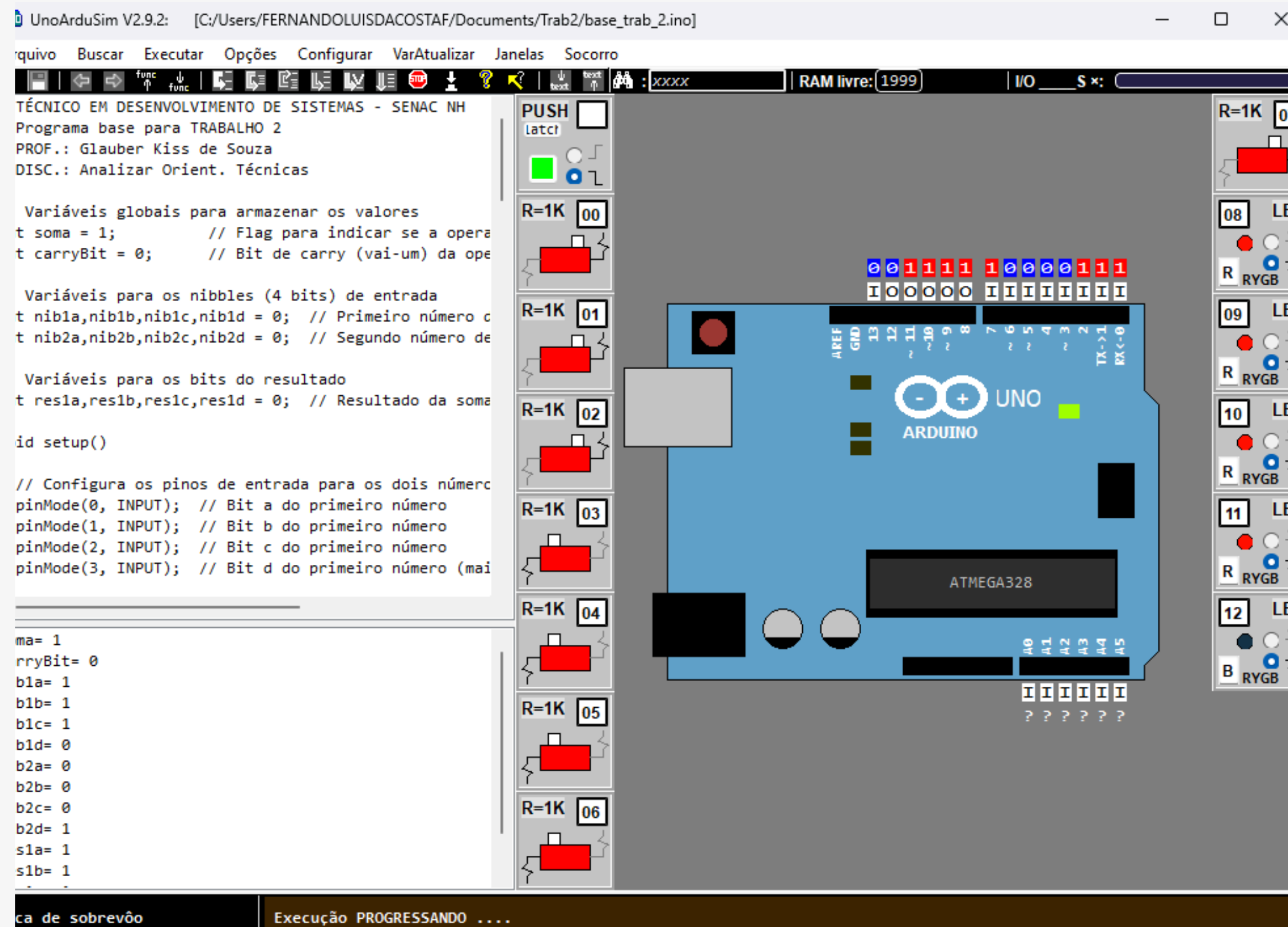
- Add binary subtraction capability
- Implement two's complement for negative numbers
- Add overflow detection circuitry
- Create visual indicators for input/output states

Medium-term Improvements

- Expand to 8-bit or 16-bit operation
- Add hexadecimal display output
- Implement seven-segment display interface
- Create a library for reusable binary operations

Long-term Vision

- Develop a complete Arithmetic Logic Unit (ALU)
- Add multiplication and division capabilities
- Create a teaching interface with serial monitor feedback
- Develop a PCB version for standalone operation



Bug Tracking

Resolved Issues

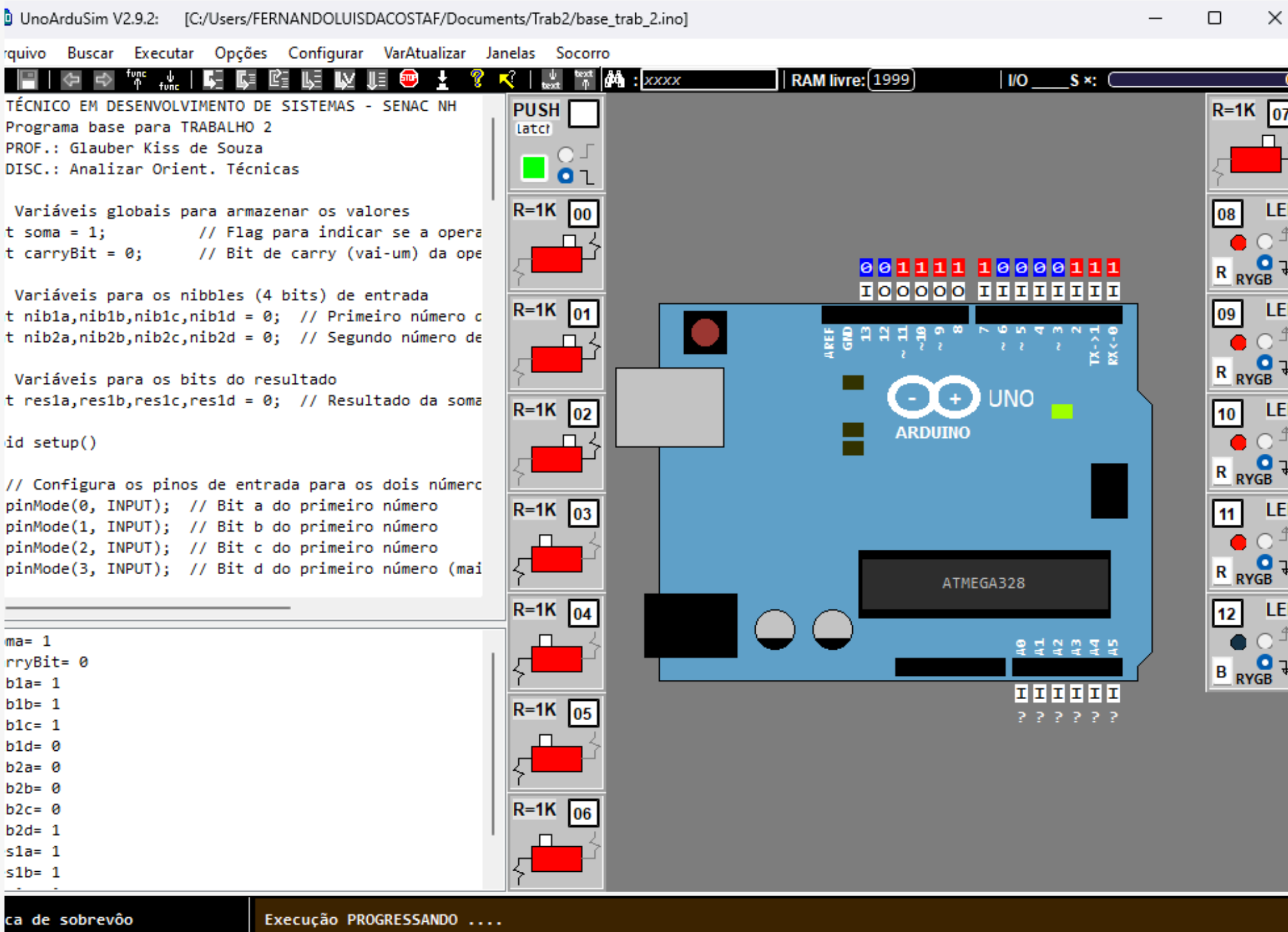
- Incorrect variable initialization - Fixed by properly initializing all variables
- Carry bit propagation error - Corrected the carry calculation logic
- Pin mode configuration - Fixed input/output pin configuration
- Bit order confusion - Clarified LSB/MSB positioning in code comments

Known Issues

- No input debouncing - Switch inputs may need debouncing for stable reading
- Limited error handling - No validation for invalid input states
- No visual feedback - Currently lacks LED indicators for input states

Issues to Address

- Implement input debouncing - Add software debouncing for switch inputs
- Add input validation - Implement checks for valid binary inputs
- Create test suite - Develop comprehensive testing procedures
- Improve documentation - Add detailed comments for educational purposes





thank you